

LABORATORY 5**CURRENT MIRROR AND DIFFERENTIAL OPAMP****OBJECTIVES**

No.	Objectives	Requirements
1	Get knowledge about the current mirror topology	- Consider the function of the current mirror topology - Knowing how to improve the precision of the copied current
2	Know the steps to design a differential opamp	- Writing the DCOP measurement functions to consider the operation of NMOS and PMOS in design - Design a basic differential opamp with the current source load - Knowing how to improve the gain of amplifier topology
3	Design a unity gain buffer	- Knowing how to analyze the feedback system - Design and verify a two-stage opamp with Miller compensation technique

LAB 5 INFORMATION

- Consider *NMOS nmos1v_lvt device* and power supply $v_{dd} = 1.0V$ for Experiment 1.
- Choose *normal, low or high threshold voltage devices* for designing a differential opamp in Experiment 2 and Experiment 3.
- Note:** All figures in the document are for reference only.



EXPERIMENT 1

1.1. Objective: Build a testbench to consider the working of the current mirror topology.

1.2. Requirements: Simulate with dc analysis.

1.3. Instruction: Assemble the circuit as shown in **Figure 1**, setting parameters for power supplies (using **vdc** source), with $L_{n,unit} = 45nm$, $W_{n,unit} = 120nm$, $vdd = 1.0V$

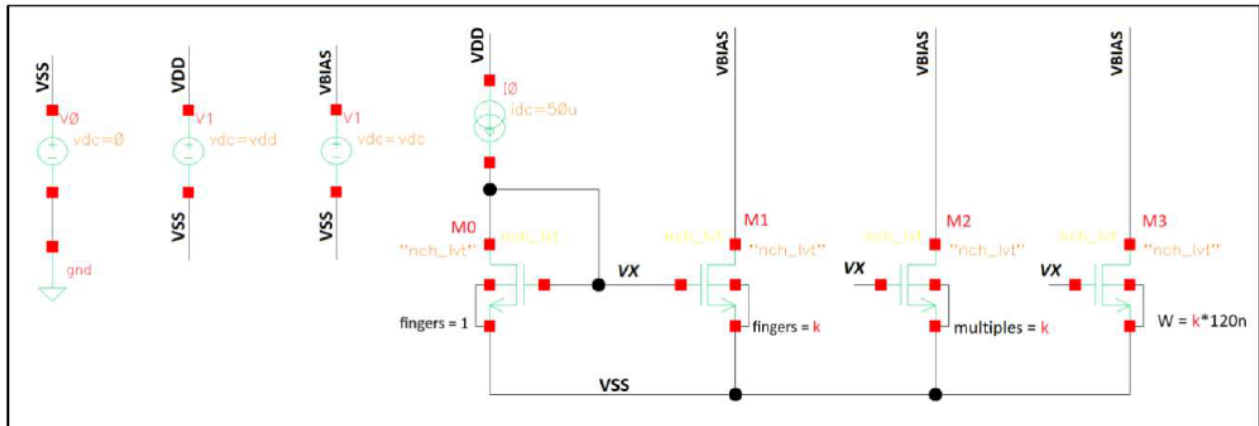


Figure 1. Schematic of the current mirror

1.4. Analyze:

➤ **1.4.1. Testcase**

Parameter	Value
Device	nmos1v_lvt
Process	TT
Temp	25°C
V _{DD}	1 V
Observe	Sweep VIBAS from 0 to VDD Copied current at M0 with k from 1 to 5

- Consider the copied current with the variation of I_{M1} , I_{M2} and I_{M3} with the voltage **VBIAS** sweep from 0V to 1.0V and sweep $k = \{1, 2, 3, 4, 5\}$

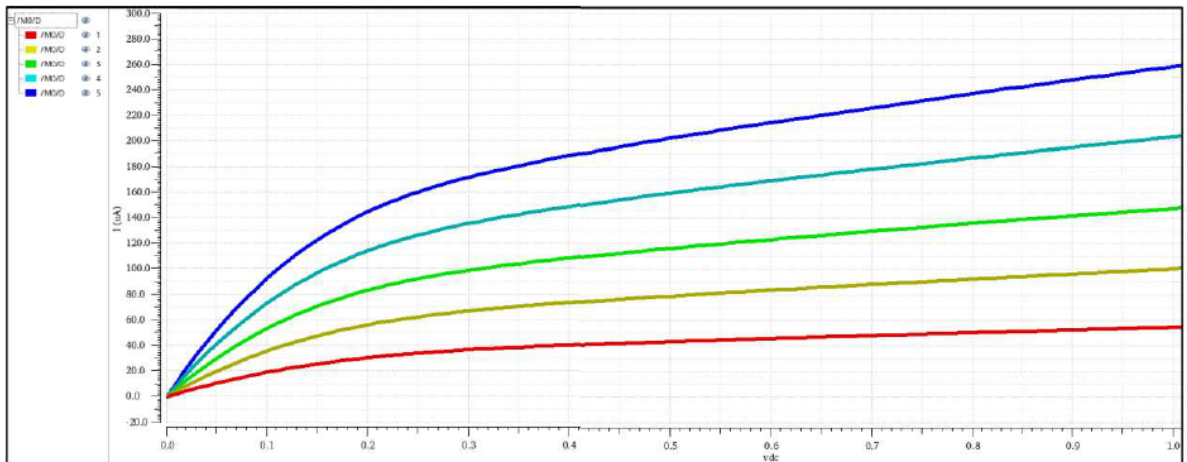
➤ 1.4.2. Waveform:

Figure 2. Example waveform of current mirror topology

➤ 1.4.3. Observe from the waveform:

- **Complete the table with the target copied current and the VBIAS required to reach this current.**

Copied ratio (k)	1	2	3	4	5
Target copied current (uA)	50	...	...	...	...
VX (mV)	...	...	...	...	...
VBIAS (mV) when fingers = k	...	...	...	...	...
VBIAS (mV) when multiples = k	...	...	...	...	...
VBIAS (mV) when $W = k \cdot W_{\text{unit}}$	...	...	...	...	...

➤ 1.4.4. Explanation (if applicable):

EXPERIMENT 2

2.1. Objective: Verify the differential amplifier with the current source load to meet the given specification.

2.2. Requirements: Simulate the dc, ac and transient analysis with an *ADE-Explorer environment*.

2.3. Instruction: Assemble the circuit as shown in **Figure 3**, setting parameters for power supplies (using **vdc** source), with $C_{load} = 20fF$, $v_{dd} = 1.0V$, $v_{bias} = 0.48V$, $v_{in_{cm}} = 0.8V$,

- $L_n = 45nm, L_p = 45nm, W_{n,unit} = 120nm, W_{p,unit} = 120nm$
- $L_n = 200nm, L_p = 200nm, W_{n,unit} = 500nm, W_{p,unit} = 500nm$

- The requirement architecture of the differential amplifier for Experiment 2.

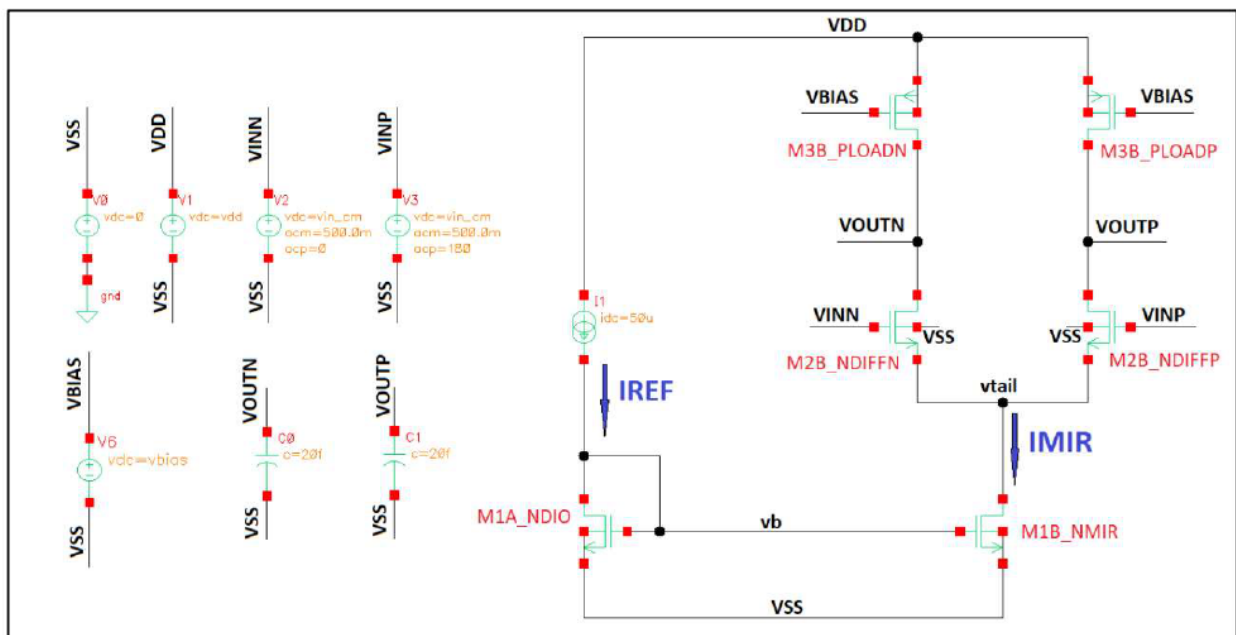


Figure 3. Schematic of differential amplifier with current source load

2.4. Analyze:➤ **2.4.1. Testcase and specification**

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25°C

V_{DD}	1 V
Reference current (IREF)	50uA
Copied bias current (IMIR)	100uA $\pm$ 3%
Capacitor load	20fF
Input common mode voltage	0.8V
Output common mode voltage	0.6V $\pm$ 3%
Bias voltage	0.48V
DC Gain	> 15dB
BW_-3dB	As much as possible

➤ 2.4.2. DC analysis:

- The requirements for the DC analysis in this amplifier are
 - All NMOS and PMOS are operated in the saturation region.
 - The error copied current between IREF and IMIR is less than 3% (means the current **IMIR is in the range [97uA, 103uA] with the IREF = 50uA.**
 - The output DC voltage **VOUTN, VOUTP** (output common mode voltage) is defined in range **from 0.582V to 0.618V (0.6V $\pm$ 3%).**

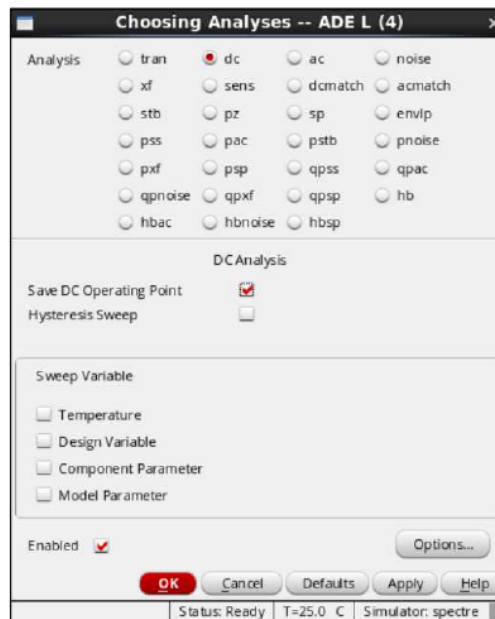


Figure 4. Setup dc analysis with saving DC Operating point option

- Using ADE-Explorer and writing the equations to consider the DCOP (DC Operating Point) of each NMOS and PMOS in design.
 - Setup dc analysis with save DC Operating point option as **Figure 4**.
 - **Figure 5** is an example of how to write the DCOP measurement equations of the M1A_NDIO device, the current IMIR, and voltage VOUTN. **Writing the DCOP measurement equations for other NMOS devices, PMOS devices and other voltage nodes in the design by yourself.**
 - Generally, the NMOS is operated in saturation when the $v_{od} = v_{gs} - v_{th} > 0$ and $v_{sat} = v_{ds} - v_{dsat} > 0$.
 - In this experiment, we try **verify the size of MOS devices having $v_{od} \geq -50mV$ and $v_{sat} \geq 0mV$ (the MOS can be operated in weak inversion).**

Name	Type	Details	EvalType	Plot	Save	Spec
vgs_M1A_NDIO	expr	OP("M1A_NDIO" "vgs")	point	☒	☐	
vth_M1A_NDIO	expr	OP("M1A_NDIO" "vth")	point	☒	☐	
vds_M1A_NDIO	expr	OP("M1A_NDIO" "vds")	point	☒	☐	
vdsat_M1A_NDIO	expr	OP("M1A_NDIO" "vdsat")	point	☒	☐	
vod_M1A_NDIO	expr	(vgs_M1A_NDIO - vth_M1A_NDIO)	point	☒	☐	> -50m
vsat_M1A_NDIO	expr	(vds_M1A_NDIO - vdsat_M1A_NDIO)	point	☒	☐	> 0m
id_MIR	expr	OP("M1B_NMIR" "ids")	point	☒	☐	range 97u 103u
VOUTN	expr	VDC("/VOUTN")	point	☒	☐	range 0.59 0.61

1. You can define the specification to the output

Figure 5. Example of equations to measure the DCOP of M1A_NDIO device

Output	Spec	Weight	Pass/Fail	Min	Max	TTNN
Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒	Filter ☒
VBIAS				480m	480m	480m
VOUTP	range 0.582 0.618		pass	589.5m	589.5m	589.5m
VOUTN	range 0.582 0.618		pass	589.5m	589.5m	589.5m
VINP				800m	800m	800m
VINN				800m	800m	800m
id_M1A_NDIO				50u	50u	50u
id_M1B_NMIR	range 97u 103u		pass	97.28u	97.28u	97.28u
vod_M1A_NDIO	> -50m		pass	-31.21m	-31.21m	-31.21m
vsat_M1A_NDIO	> 0		pass	261.5m	261.5m	261.5m
vod_M1B_NMIR	> -50m		pass	-31.14m	-31.14m	-31.14m
vsat_M1B_NMIR	> 0		pass	202.9m	202.9m	202.9m
vod_M2B_NDIFFN	> -50m		pass	-37.68m	-37.68m	-37.68m
vsat_M2B_NDIFFN	> 0		pass	208.2m	208.2m	208.2m
vod_M2B_NDIFFP	> -50m		pass	-37.68m	-37.68m	-37.68m
vsat_M2B_NDIFFP	> 0		pass	208.2m	208.2m	208.2m
vod_M3B_PLOA...	> -50m		pass	36.81m	36.81m	36.81m
vsat_M3B_PLOA...	> 0		pass	323m	323m	323m
vod_M3B_PLOA...	> -50m		pass	36.81m	36.81m	36.81m
vsat_M3B_PLOA...	> 0		pass	323m	323m	323m

Figure 6. Example of DCOP result in differential amplifier design

- **Figure 6** is an example of DCOP result that all MOS devices are operated in saturation region ($v_{od} \geq -50mV$, $v_{sat} \geq 0mV$) and **meet the specification about copied current $IMIR$ and output voltage V_{OUTN} , V_{OUTP} .**
- **Students can use stacked devices to increase the effective channel length of the MOS, as shown in Figure 7.**

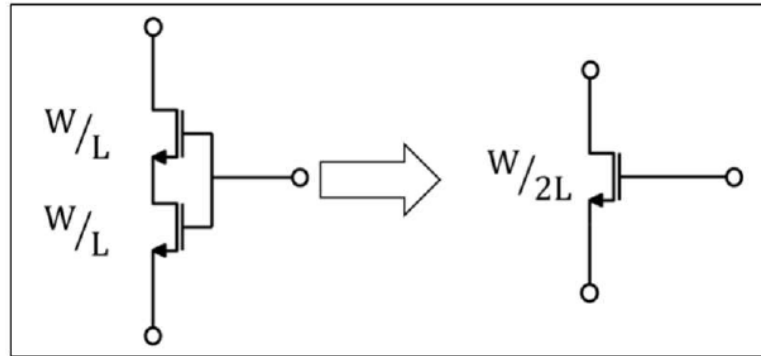


Figure 7. Stacked MOS devices

- **For DC operating point (DCOP) checks, we have to examine the devices that are furthest from the VDD/VSS sources as Figure 8.**

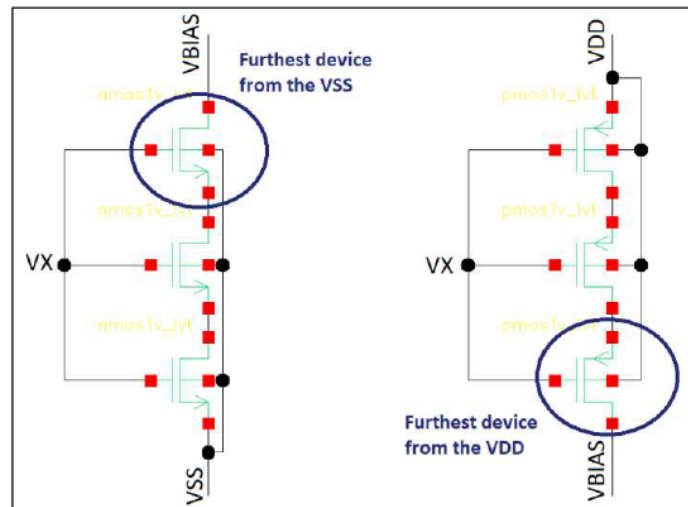


Figure 8. Stacked MOS device DCOP check rule

Device name	Size
M1A_NDIO	$L_n = 200nm$, $W_n = 500nm$, $fingers = \dots$, $multipliers = \dots$, $stack = \dots$
M1B_NMIR	$L_n = 200nm$, $W_n = 500nm$, $fingers = \dots$, $multipliers = \dots$, $stack = \dots$

M2B_NDIFFN	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$
M2B_NDIFFP	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
M3B_PLOADN	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
M3B_PLOADP	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$

➤ 2.4.3. AC analysis:

- Create the testbench as **Figure 3**, the input ac voltage VINP and VINN equal to 500mV and differential phase between two input signal is 180°.
- **Measure the Gain and -3dB BW of the differential amplifier design by writing the Calculator function.**

$$|H(\omega)| = \frac{V_{OUTP} - V_{OUTN}}{V_{INP} - V_{INN}}$$

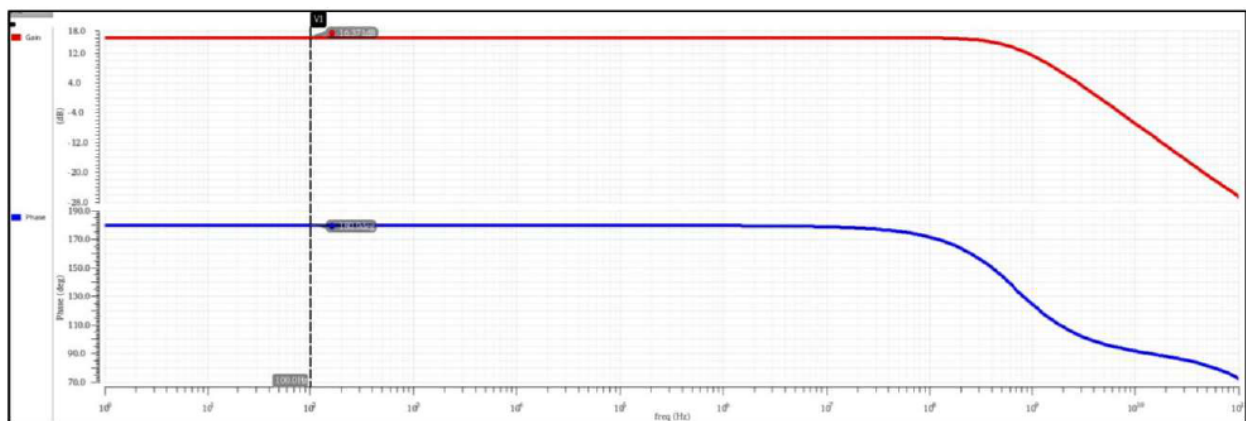


Figure 9. Example of ac analysis result

- The requirement DC Gain is **greater than 15dB**, if your design doesn't meet the gain specification. You must follow 4 steps:
 - Study how to increase the gain of amplifier, changing size (increase or decrease) of which MOS device can help increasing gain?
 - Verify size of MOS to increase gain.

- **Redo the DC analysis step** to ensure your design is meet the current and voltage specification (all MOS device must be operated in saturation region).
- Check the DC Gain with ac analysis.
- **If you still don't meet the DC Gain specification, please try verify the size of MOS devices one more time.**

Parameter	Value
Gain (dB)	...
BW_-3dB (Hz)	...

➤ 2.4.4. TRAN analysis:

- Build the testbench to simulate the differential amplifier with transient analysis.
- Choose amplitude of sinuous differential input signal is 10mV, frequency = 1MHz.
- **Calculate the gain at low frequency and compare with AC analysis. Explain the waveform.**

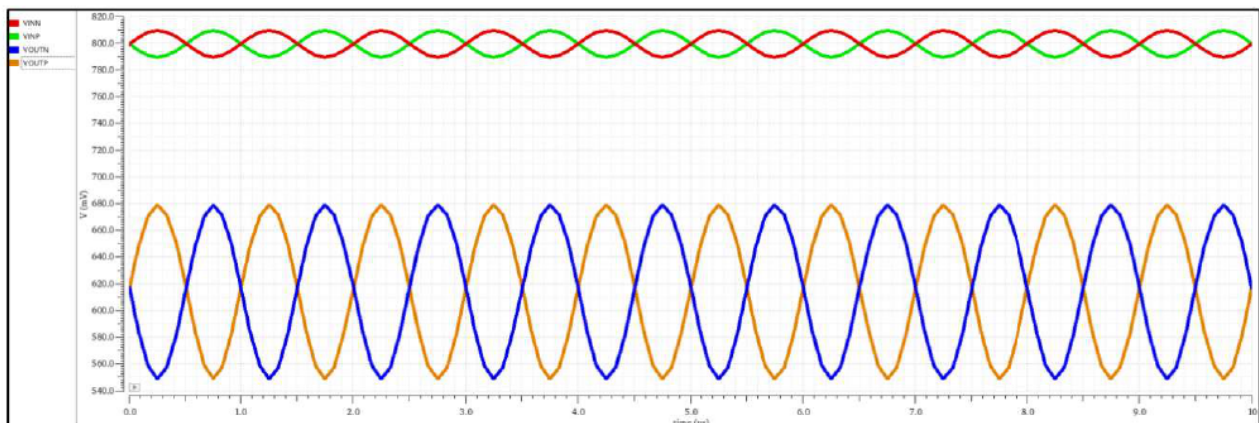


Figure 10. Example of transient analysis result

- **Increase the amplitude of sinuous differential input signal to 100mV, frequency = 1MHz. Explain the output waveform result.**

EXPERIMENT 3

3.1. Objective: Verify the unity gain buffer with two-stage miller operational amplifier topology

3.2. Requirements: Simulate the dc, ac and transient analysis with an **ADE-Explorer environment**.

3.3. Instruction: Assemble the circuit as shown in **Figure 14** and **Figure 15**, with $v_{dd} = 1.0V$, $v_{in_{cm}} = 0.8V$.

- $L_n = 45nm, L_p = 45nm, W_{n,unit} = 120nm, W_{p,unit} = 120nm$
- $L_n = 200nm, L_p = 200nm, W_{n,unit} = 500nm, W_{p,unit} = 500nm$

- The simplest operational amplifier is the simple differential amplifier studied earlier. This amplifier can be improved by adding a second stage as an inverting amplifier with a current source load. The two-stage amplifier shown in **Figure 11** is referred to as a Miller Opamp.

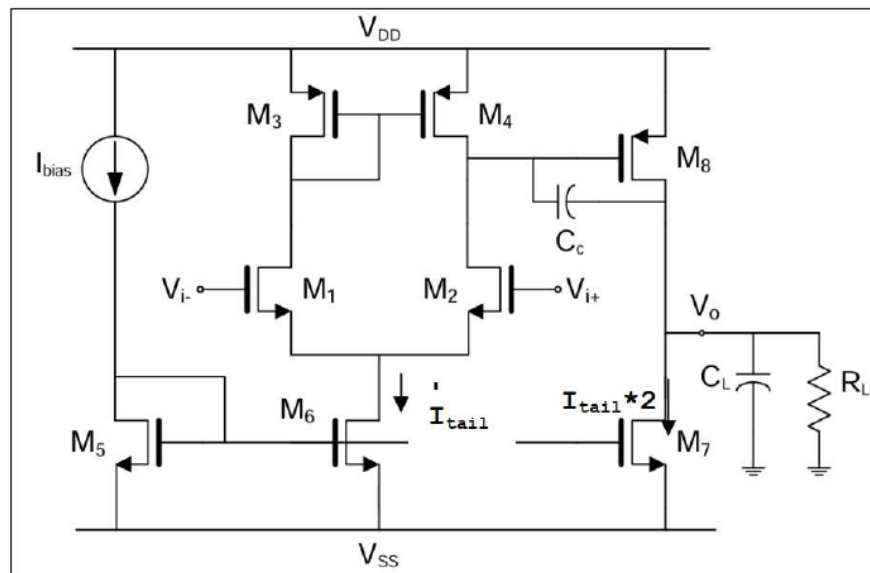


Figure 11. Two-stage miller operational amplifier

- The Miller opamp has a low frequency gain of:

$$A_v = G_m \cdot (R_{out} \parallel R_L)$$

- The transconductance is given by:

$$G_m = \frac{g_{m1,2}}{g_{02} + g_{04}} \cdot g_{m8}$$

- The output resistance is given by:

$$R_{out} = r_{07} \parallel r_{08}$$

- The frequency response of an operational amplifier will be analyzed using the macro-model of the opamp shown in **Figure 12**. The capacitor C_{in} models the input capacitance of the opamp, which is mostly gate to source capacitance. The sub-circuit consisting of G_{mA} , R_A and C_A model the gain and frequency response of the input stage. The capacitance C_A includes the input capacitance of the second stage and the output capacitance of the first stage. The components G_{mB} , R_B and C_B model the second stage. The load capacitor and resistor are also included in R_B and C_B .

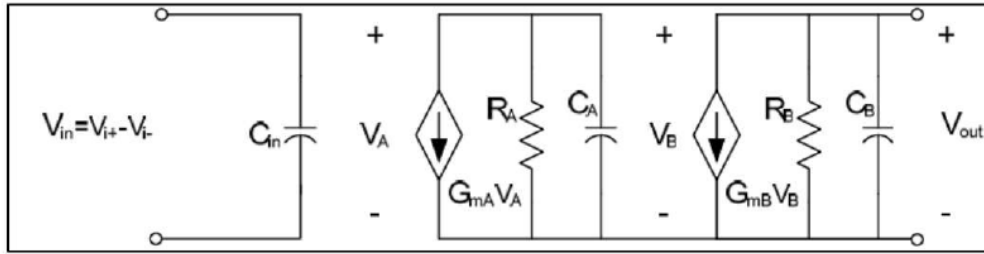


Figure 12. Operational Amplifier Macro-Model

- The transfer function of the macro-model is given by:

$$H(s) = \frac{A_{v0} p_1 p_2}{(s + p_1)(s + p_2)} = \frac{\frac{G_{mA} G_{mB}}{C_A C_B}}{\left(s + \frac{1}{R_A C_A}\right) \left(s + \frac{1}{R_B C_B}\right)}$$

- This transfer function assumes zero source resistance. Notice the two poles are approximately equal. The capacitors C_A and C_B are dominated by gate to source capacitances, and R_A and R_B are the parallel connected small-signal drain to source resistances.
- Due to the poles being located close together and the large DC gain, the system is unlikely to be stable in unity-gain feedback configuration, therefore some form of compensation is required. The modified macro model shown in **Figure 13** uses capacitor C_C to compensate the frequency response of the opamp by splitting the two poles.

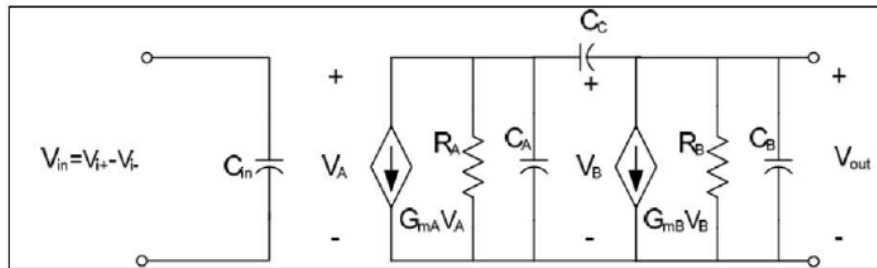


Figure 13. Operational Amplifier Macro-Model with Compensation Capacitor C_C

- Assuming R_A is large ($R_A \approx R_B$ and $R_A \gg 1/G_{mB}$) and C_A is small ($C_B, C_C \gg C_A$). The transfer function for the operational amplifier with the compensation capacitor is:

$$H(s) = \frac{\frac{A_{v0}p_1p_2}{z}(s - z)}{(s + p_1)(s + p_2)} = \frac{\frac{G_{mA}}{C_B}\left(s - \frac{G_{mB}}{C_C}\right)}{\left(s + \frac{1}{G_{mB}R_AR_BC_C}\right)\left(s + \frac{G_{mB}}{C_B}\right)}$$

- With the transfer function in factored form, we can find the open loop DC gain, poles and zero of the compensated opamp. They are given by:

$$A_v = G_{mA}G_{mB}R_AR_B$$

$$p_1 = \frac{-1}{G_{mB}R_BC_C R_A}$$

$$p_2 = \frac{-G_{mB}}{C_B}$$

$$z = \frac{G_{mB}}{C_C}$$

- Note that the addition of the compensation capacitor C_C caused the poles to split. One pole moved closer to the origin by a factor of $G_{mB}R_B$, while the other pole moved away from the origin by a factor of $G_{mB}R_B$. This compensation technique is called "pole splitting".
- To assure the feedback system is stable at unity gain ($\beta = 1$), the phase margin must be examined.

$$PM = 180^\circ - \tan^{-1}\left(\frac{\omega_{0dB}}{p_1}\right) - \tan^{-1}\left(\frac{\omega_{0dB}}{p_2}\right) - \tan^{-1}\left(\frac{\omega_{0dB}}{z}\right)$$

$$PM \approx 90^\circ - \tan^{-1}\left(\frac{\omega_{0dB}}{p_2}\right) - \tan^{-1}\left(\frac{\omega_{0dB}}{z}\right)$$

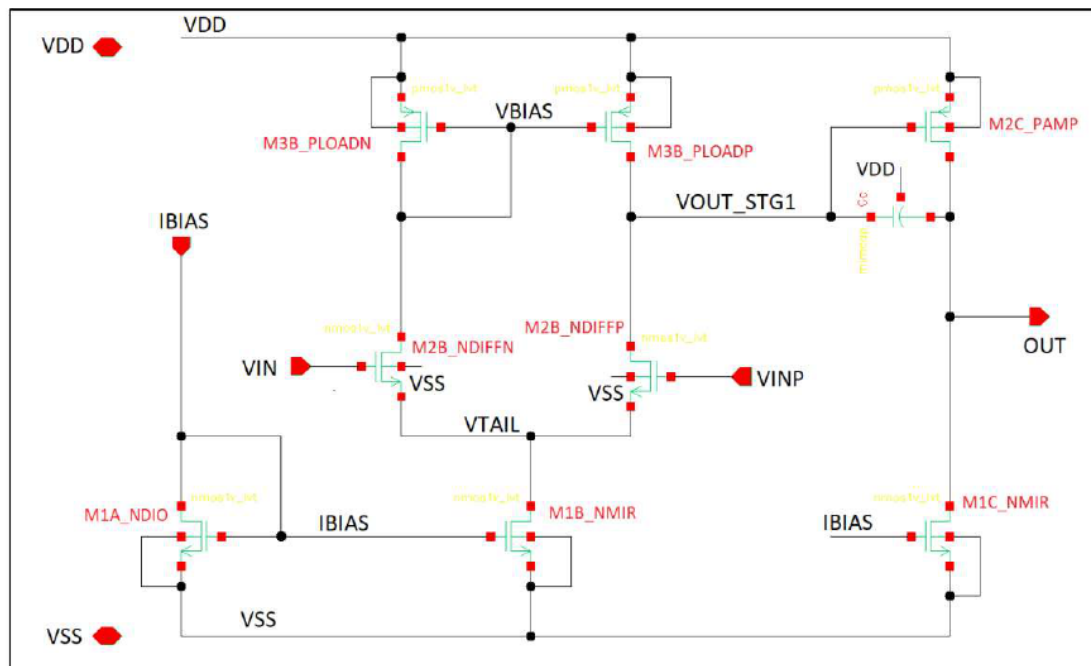
- The phase margin is improved by moving the non-dominant pole and zero to higher frequencies away from the unity gain frequency. The phase margin can also be improved by using compensation techniques which place the zero in the left half plane.
- The slew rate is determined by the compensation capacitance and the tail current:

$$SR = \frac{I_{tail}}{C_C}$$



3.4. Analyze:➤ **2.4.1. Testcase and specification**

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25°C
V _{DD}	1 V
Reference current (I _{bias})	50uA
Copied bias current 1stg (I _{tail})	100uA±3%
Copied bias current 2stg (I _{tail} *2)	> 200uA–3% or >194uA
Capacitor load	20fF
Input common mode voltage	0.8V
V _{od} (DCOP analysis)	> –50mV
V _{sat,margin} (DCOP analysis)	> 0mV
DC STB Gain (STB analysis)	> 30dB
f _{UGB} (STB analysis)	> 500MHz
Phase margin (STB analysis)	> 60deg

**Figure 14.** Schematic of 2-stage opamp

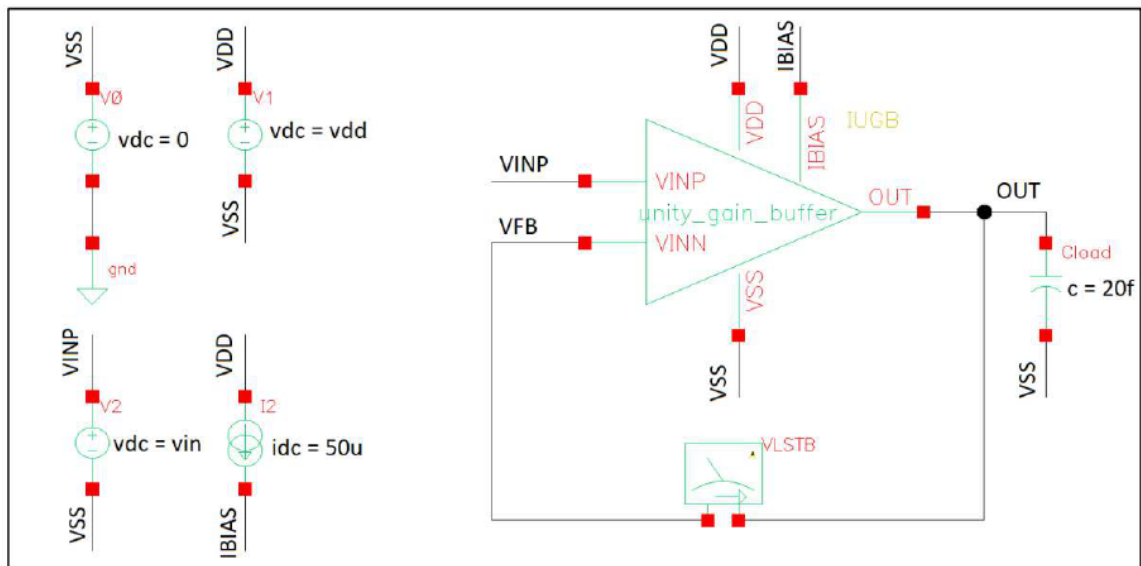


Figure 15. The unity gain buffer testbench for DC and STB analysis

➤ 2.4.2. DC analysis:

- Set up a testbench as shown in **Figure 15** and verify the operation of a **unity-gain buffer** using a **two-stage Miller operational amplifier** topology.
- Students must **verify the DC operating point with the closed-loop configuration** as **Figure 15**.

Parameter						TTNN
gpd045.scs						tt
temperature						25
vdd						1
Output	Spec	Weight	Pass/Fail	Min	Max	TTNN
Filter	Filter	Filter	Filter	Filter	Filter	Filter
VOUT				798.3m	798.3m	798.3m
VINP				800m	800m	800m
VINN				798.3m	798.3m	798.3m
id_M1A_NDIO				50u	50u	50u
id_M1B_NMIR	range 97u 103u		pass	97.09u	97.09u	97.09u
id_M1C_NMIR	> 194u		pass	256.2u	256.2u	256.2u
vod_M1A_NDIO	> -50m		pass	-31.2m	-31.2m	-31.2m
vsat_M1A_NDIO	> 0		pass	261.5m	261.5m	261.5m
vod_M1B_NMIR	> -50m		pass	-31.14m	-31.14m	-31.14m
vsat_M1B_NMIR	> 0		pass	199.1m	199.1m	199.1m
vod_M2B_NDIFFN	> -50m		pass	-36.09m	-36.09m	-36.09m
vsat_M2B_NDIFFN	> 0		pass	176.2m	176.2m	176.2m
vod_M2B_NDIFFP	> -50m		pass	-34.82m	-34.82m	-34.82m
vsat_M2B_NDIFFP	> 0		pass	165.3m	165.3m	165.3m
vod_M3B_PLOA...	> -50m		pass	-37.13m	-37.13m	-37.13m
vsat_M3B_PLOA...	> 0		pass	380.3m	380.3m	380.3m
vod_M3B_PLOA...	> -50m		pass	-37.09m	-37.09m	-37.09m
vsat_M3B_PLOA...	> 0		pass	390.9m	390.9m	390.9m
vod_M1C_NMIR	> -50m		pass	-30.13m	-30.13m	-30.13m
vsat_M1C_NMIR	> 0		pass	684.2m	684.2m	684.2m
vod_M2C_PAMP	> -50m		pass	-27.44m	-27.44m	-27.44m
vsat_M2C_PAMP	> 0		pass	133.4m	133.4m	133.4m

Figure 16. Example of DCOP result in unity gain buffer design

Device name	Size
M1A_NDIO	$L_n = 200nm, W_n = 500nm,$ $fingers = ..., multipliers = ..., stack = ...$
M1B_NMIR	$L_n = 200nm, W_n = 500nm,$ $fingers = ..., multipliers = ..., stack = ...$
M2B_NDIFFN	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$
M2B_NDIFFP	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
M3B_PLOADN	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
M3B_PLOADP	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
M1C_NMIR	$L_n = 200nm, W_n = 500nm,$ $fingers = ..., multipliers = ..., stack = ...$
M2C_PAMP	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$
Miller cap Cc (mimcap)	$W = 1\mu m, L = 1\mu m$ $multipliers = ...$ (cap value = ... fF)

➤ 3.4.3. STB analysis:

- Students **set up and use STB analysis for frequency-response simulation** (not AC analysis) as **Figure 17**.

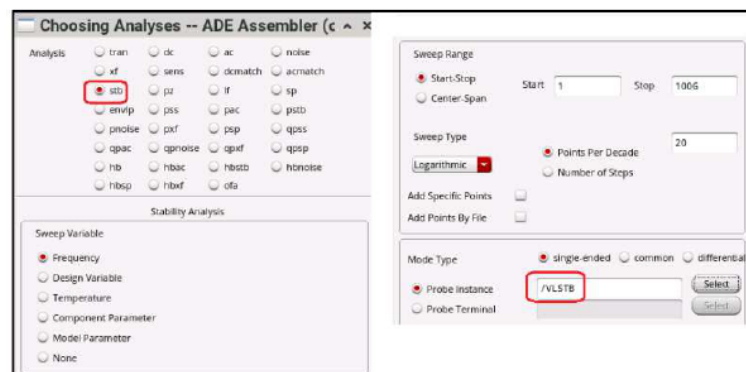


Figure 17. STB setup

- Write the measurements as shown in **Figure 18** to measure DC gain, Phase margin, and Gain margin.

Loop_Gain_Phase	expr	phaseDegUnwrapped(getData("loopGain" ?result "stb"))
Loop_Gain_dB20	expr	db(mag(getData("loopGain" ?result "stb")))
Gain_DC	expr	value(Loop_Gain_dB20 1000)
Phase_Margin	expr	getData("phaseMargin" ?result "stb_margin")
Gain_Margin	expr	getData("gainMargin" ?result "stb_margin")
BW_UGB	expr	cross(Loop_Gain_dB20 "0" 1 "falling" nil nil nil)

Figure 18. STB measurement**Figure 19.** STB result

Parameter	Value
DC Gain	
Phase Margin	
Gain Margin	
f_{UGB}	

➤ **3.4.4. TRAN analysis:**

- Simulate the transient response with a pulse signal of 100mVpp at 100KHz, with a 0.8V offset as shown in **Figure 20**, and analyze the output response.

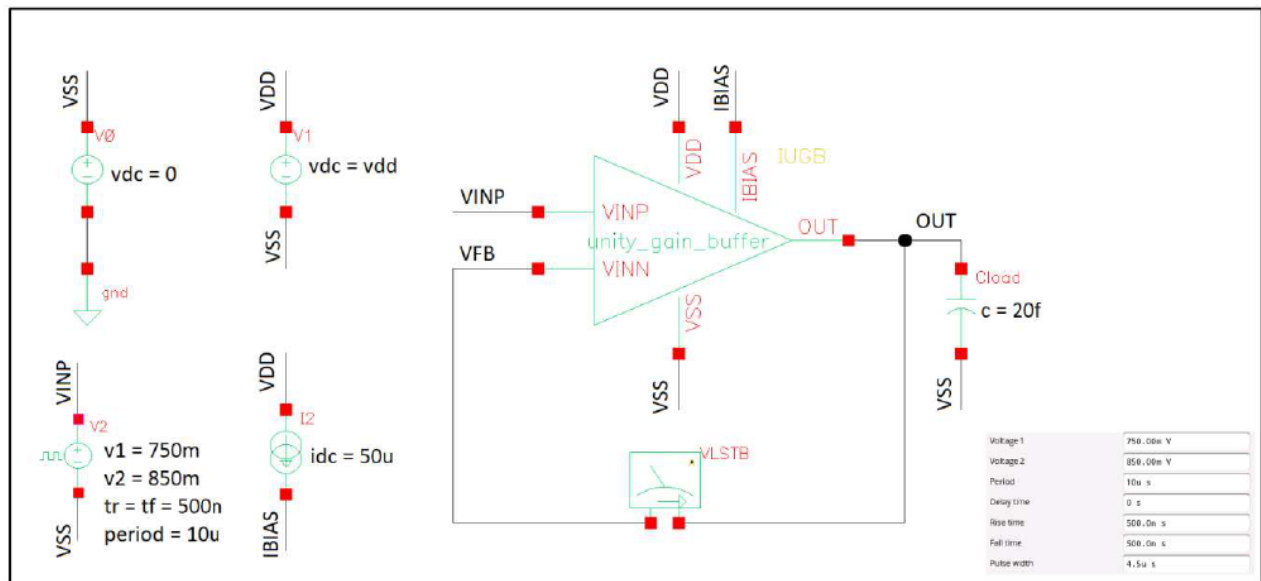


Figure 20. The unity gain buffer testbench for TRAN analysis

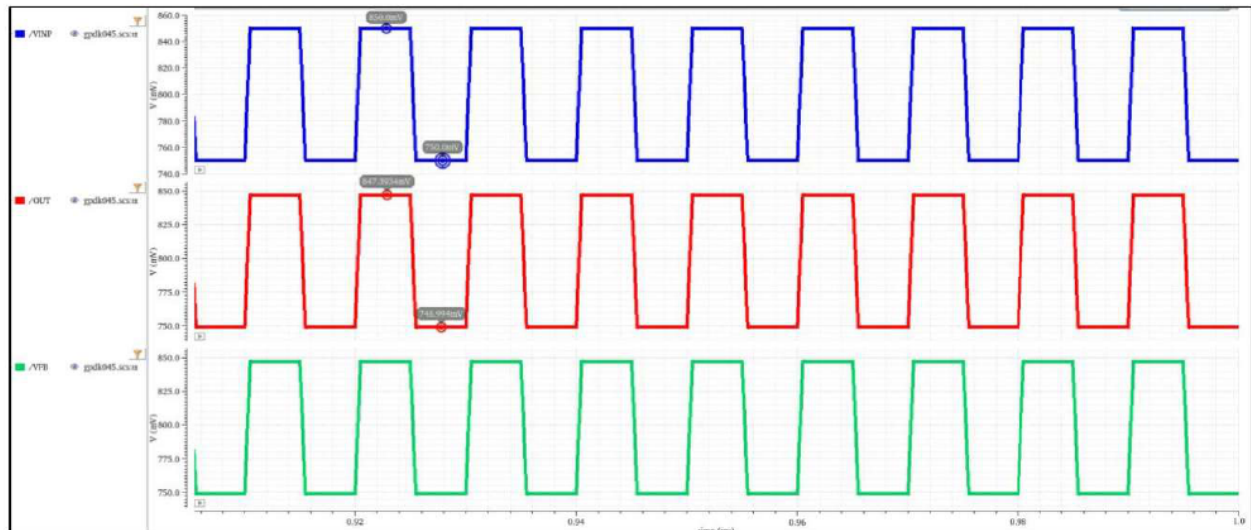


Figure 21. The result of TRAN analysis

- **Figure 21** shows the result of the transient analysis with a pulse response. The input voltage is **100mVpp**, but the buffer output is **98.4mVpp**, a drop of **1.6mVpp**.

Parameter	Value
Input voltage	100mVpp
Output voltage	...Vpp
Voltage drop	...Vpp